

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	MS-MSE
Program Name	Master of Science in Materials Science and Engineering
Assessment Year	2025
Program Assessment Coordinator	Heng Wang
Year of Current Assessment Plan	2025
No Assessment This Year per Plan	☒
LO(s) Assessed	LO2 (collecting and archiving data only)
Presented at program authority meeting date: <i>Please attach minutes</i>	TBD

Methodology

LO#1

- ≠ Learning goal assessed: Graduates should have a thorough understanding of the relationships between materials, manufacturing, and product properties.
- ≠ Sample size: All students completing MMAE 588
- ≠ Semesters in which artifacts were collected: Fall, Spring
- ≠ Artifact: For each course, performance of the students in 4 questions chosen from a pool of questions specifically designed to assess this learning goal. The questions are delivered to the students in the last third of the semester. Each student will be scored out of 4.
- ≠ Proficiency ratings
 - ⌞ scores of >75% points indicate **P** proficient.
 - ⌞ scores of >55% points indicate **M** meets expectations.
 - ⌞ scores below 55% points indicate **D** does not meet expectations.

LO#2

- ≠ Learning goal assessed: Graduates should be able to solve materials related numerical problems.
- ≠ Sample size: All students completing a subset of MMAE 554
- ≠ Semesters in which artifacts were collected: Fall, Spring
- ≠ Artifact: For each course, performance of the students in 4 questions chosen from a pool of questions especially designed to assess this learning goal. The questions are delivered to the students in the last third of the semester. Each student will be scored out of 4.
- ≠ Proficiency ratings

- ⚡ scores of >75% points indicate **P** proficient.
- ⚡ scores of >55% points indicate **M** meets expectations.
- ⚡ scores below 55% points indicate **D** does not meet expectations.

LO#3

- ⚡ Learning goal assessed: For graduates doing thesis, they will have the ability to conduct research, publish scientific findings, and disseminate that knowledge.
- ⚡ Sample size: All students completing their MS thesis.
- ⚡ Semesters in which artifacts were collected: Fall, Spring
- ⚡ Artifact: A specially designed questionnaire filled out by the thesis committee after the thesis defense exams. The committee members come to a consensus and rate the students on a scale of 1-5 on each of the following:
 - ⚡ Research problem is clearly described
 - ⚡ Literature and context of the work are demonstrated
 - ⚡ Solution approach, methods and/or tools are described
 - ⚡ Results are analyzed, interpreted and communicated well
 - ⚡ Written and oral communication are effective
 - ⚡ Critical thinking consistent with an independent researcher
- ⚡ Proficiency ratings
 - ⚡ scores of 4 points and above for all items indicate **P** proficient.
 - ⚡ scores of 3 points and above for all items indicate **M** meets expectations.
 - ⚡ scores below 3 points on any item indicate **D** does not meet expectations.

Results

Artifact samples

Artifact for LO2 from MMAE 554. Four questions from a bank of five were assigned as a quiz. Sample scores listed below. "C" for correct, "I" for incorrect.

1

Multiple Choice 1 point

Estimate the carrier concentration in a metal that has a conductivity of $\sigma = 10^5$ S/cm, and relaxation time $\tau = 1 \times 10^{-14}$ s, take effective mass of electrons $m^* = 0.5 m_e$

☐ $1.78 \times 10^{26} \text{ cm}^{-3}$

☒ $1.78 \times 10^{22} \text{ cm}^{-3}$

☐ $1.78 \times 10^{21} \text{ cm}^{-3}$

☐ $8.9 \times 10^{21} \text{ cm}^{-3}$

MS-MSE LO2

2

Multiple Choice 1 point

A semiconductor has an electron concentration of 10^{16} cm^{-3} and a hole concentration of 10^{14} cm^{-3} . If the electron mobility is $1400 \text{ cm}^2/\text{V}\cdot\text{s}$ and the hole mobility is $450 \text{ cm}^2/\text{V}\cdot\text{s}$, what is the electrical conductivity of the material? ($e = 1.6 \times 10^{-19} \text{ C}$)

☒ 2.24 S/cm

☐ 224 S/cm

☐ 22.4 S/cm

☐ 2240 S/cm

MS-MSE LO2

3

Multiple Choice 1 point

What is the estimated voltage response, of a pyroelectric motion sensor, made of a PZT ceramic film of 0.1mm thick, to a temperature change of 1mK? (the relative dielectric constant for PZT is 380, dielectric constant of vacuum is $8.85 \times 10^{-12} \text{ C}\cdot\text{V}^{-1}\text{m}^{-1}$, The pyroelectric coefficient is $380 \times 10^{-6} \text{ C}\cdot\text{m}^{-2}\text{K}^{-1}$)

☐ 1.5 mV

☐ 150 μV

☒ 15 mV

☐ 1.5 μV

MS-MSE LO2

4

Multiple Choice 1 point

The intensity of a 589nm light passing through a gold film was measured to be about 15% of the initially transmitted light. What is the thickness of the gold film? (the damping constant for gold at this wavelength is 3.2)

☐ 18 nm

☒ 28 nm

☐ 180 nm

☐ 140 nm

MS-MSE LO2

	Sample 1		
P	C, C, C, C (100%)		
M	C, I, C, C (75%)		
D			

Description of findings

Due to insufficient sample size of students in this program. The department has decided to only collect and archive artifacts from students. Analysis will be performed when sufficient data is collected.

Continuous Improvement, CI, plans

MS-MSE program curriculum has undergone major changes over the past year. The MSE faculty will monitor student outcomes of the current curriculum as well as student exit surveys. They meet to discuss improvement plans. Based on student body size this will be every three years.

Prior CI plan impact

Also, please refer to the previous assessment report for the LOs assessed. Were any CI plans recommended? Were they implemented? Did the plans work (were improvements seen in this cycle)?

☒ no previous CI plans available

This cycle's CI plan

Describe any CI plans developed from this cycle's data, with

- Specific curricular or instructional changes in response to this assessment data,
- A timeline for implementation,
- As well a description of who will participate in and manage these plans

☐ After consideration and presentation to the program authority, no CI plans are recommended.

MS-MSE has an extremely small student body (1-3 students per year). Moving forward, the faculty will collect artifacts from more courses (updated assessment plan attached). Even so, the faculty expect to conduct a full assessment once every 4-5 years in the future.